

Part 4(b): Value error versus iteration

Error measure. Backward DP (Part 1) gives the exact optimal cost-to-go J_{DP} on every augmented state. It is therefore used as the reference, and the error of any value table J is measured in the sup-norm over all states and stages,

$$\varepsilon(J) = \|J - J_{\text{DP}}\|_{\infty} = \max_{z \in \mathcal{Z}} |J(z) - J_{\text{DP}}(z)|, \quad (1)$$

where $\mathcal{Z}$ contains the 3185 states $z = (k, e, \eta)$, $k = 0, \dots, 12$, together with the absorbing state Δ . For value iteration (VI) the error is recorded after every synchronous sweep $J_s = TJ_{s-1}$. For policy iteration (PI) it is recorded after every exact policy evaluation, i.e. $\varepsilon(J_{\mu_i})$ for the policy μ_i of round i .

Results. Table ?? and Figure ?? show the error history. Exact zeros cannot be drawn on a logarithmic axis, so they are plotted at 10^{-16} .

Iteration	1	2	3	4	5	6	7
VI	8830	7530	6090	4680	3420	2360	1810
PI	1850	18.2	8.13	0.852	0	–	–
Iteration	8	9	10	11	12	13	14
VI	1340	939	432	124	12.5	0	0

Table 1: Maximum value error $\|J - J_{\text{DP}}\|_{\infty}$ after each VI sweep and after each PI policy evaluation.

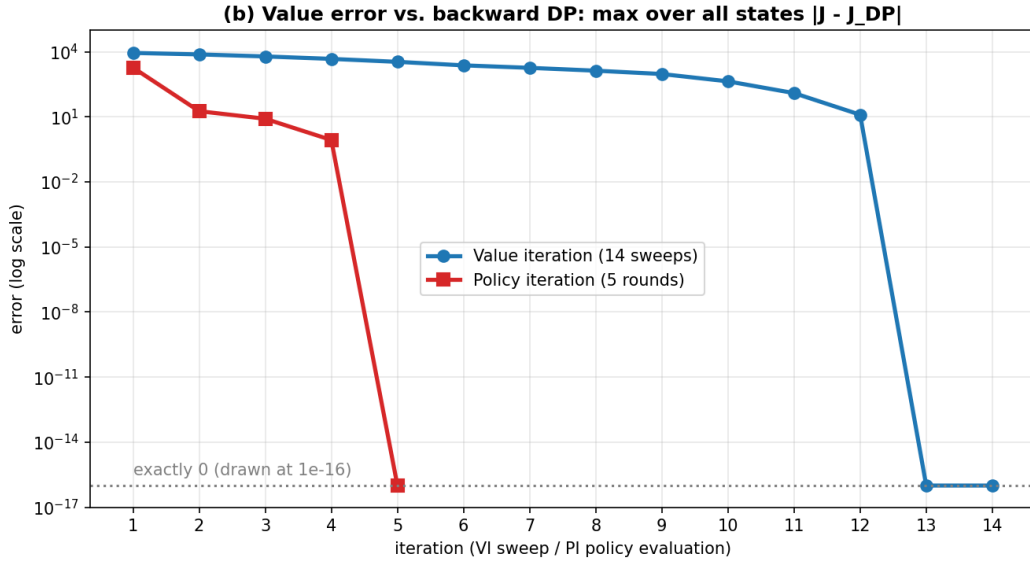


Figure 1: Value error relative to backward DP versus iteration (logarithmic scale). VI: error after each sweep. PI: error after each policy evaluation.

Interpretation of the VI curve. Starting from $J_0 \equiv 0$, sweep 1 makes layer $k = 12$ exact ($J = g_N$), and sweep s makes layers $k = 13 - s, \dots, 12$ exact, since each layer is computed from

the already exact layer behind it. The error is a maximum over *all* states, so it stays large until the last layer $k = 0$ is exact. This happens at sweep 13, where the error becomes exactly 0; sweep 14 only confirms $TJ = J$ (Bellman residual $0 < 10^{-10}$). Because $J_0 \equiv 0$ and all costs are nonnegative, VI approaches J^* from *below*: its early values account for only the last few stages of cost. For example, after sweep 1 the state $(9, -33, -2)$ has value $33^2 + 0.25 \cdot 4 + 0.5 = 1090.5$, while $J^*(9, -33, -2) = 9919.99$, which gives the error ≈ 8830 .

Interpretation of the PI curve. Each PI value table is the exact cost of a feasible policy, so $J_{\mu_i} \geq J^*$ and PI approaches J^* from *above*. The initial policy $u = -\text{sign}(\eta)$ reacts only to the drift and ignores the reference, so its error is largest where a position correction is needed but no drift exists, e.g. $J_{\mu_1}(3, 15, 0) = 4366.4$ versus $J^*(3, 15, 0) = 2517.35$ (error ≈ 1850). By the policy improvement theorem, $J_{\mu_{i+1}} \leq J_{\mu_i}$ at every state, and the error drops sharply ($1850 \rightarrow 18.2 \rightarrow 8.13 \rightarrow 0.852 \rightarrow 0$). It reaches zero after the fifth evaluation, when no control changes.

Why fewer iterations does not imply less computation. PI needs 5 iterations and VI needs 14, but an iteration is not a fixed unit of work. Let one Q -evaluation denote the basic operation $Q(z, u) = g(e, \eta, u) + \sum_w p(w) J(k+1, e + \eta - \Delta r_k, \eta + u + w)$. Since $|\mathcal{U}(\eta)| = 1, 2, 3, 2, 1$ for $\eta = -2, \dots, 2$, a full Bellman backup over the $552 \cdot 5 = 2760$ decision states ($k \leq 11$) costs $9 \cdot 552 = 4968$ Q -evaluations. A policy evaluation uses only the fixed control and costs 2760.

Method	Work per iteration	Iterations	Total Q -evaluations	Time (ms) [†]
Backward DP	4968	1	4,968	26.1
Value iteration	4968	14	69,552	237.2
Policy iteration	$2760 + 4968 = 7728$	5	38,640	89.6

Table 2: Computational work. [†]Median of 7 runs, plotting excluded; absolute times are machine dependent.

Three observations follow.

1. **Iterations have different costs.** One PI round (evaluation plus improvement) costs $7728/4968 \approx 1.6$ VI sweeps, so the 5 PI rounds correspond to about 7.8 sweeps of work, not 5. PI still wins here, but by a much smaller factor than $14/5$ suggests.
2. **Exploiting structure matters most.** Backward DP uses the finite-horizon ordering directly and computes each layer exactly once. With a single iteration it does about $14\times$ less work than VI and $8\times$ less than PI. The measured times follow the same ranking.
3. **Evaluation can be expensive in general.** Here exact policy evaluation is one backward pass because the horizon is finite. For infinite-horizon (e.g. discounted) problems, exact evaluation requires solving a linear system $J_\mu = g_\mu + \alpha P_\mu J_\mu$ with one unknown per state, which costs $O(n^3)$ for n states. A single PI iteration may then be far more expensive than many VI sweeps.

Hence the iteration count alone is not a measure of computational cost; the fair comparison is total work, i.e. (number of iterations) $\times$ (cost per iteration), or measured run time.